

Roxarsone, Inorganic Arsenic, and Other Arsenic Species in Chicken: A U.S.-Based Market Basket Sample



Background: Inorganic arsenic (iAs) causes cancer and possibly other adverse health outcomes. Arsenic-based drugs are permitted in poultry production; however, the contribution of chicken consumption to iAs intake is unknown. Objectives: We sought to characterize the arsenic species profile in chicken meat and estimate bladder and lung cancer risk associated with consuming chicken produced with arsenic-based drugs. Methods: Conventional, antibiotic-free, and organic chicken samples were collected from grocery stores in 10 U.S. metropolitan areas from December 2010 through June 2011. We tested 116 raw and 142 cooked chicken samples for total arsenic, and we determined arsenic species in 65 raw and 78 cooked samples that contained total arsenic at ≥ 10 $\mu\text{g/kg}$ dry weight. Results: The geometric mean (GM) of total arsenic in cooked chicken meat samples was 3.0 $\mu\text{g/kg}$ (95% CI: 2.5, 3.6). Among the 78 cooked samples that were speciated, iAs concentrations were higher in conventional samples (GM = 1.8 $\mu\text{g/kg}$; 95% CI: 1.4, 2.3) than in antibiotic-free (GM = 0.7 $\mu\text{g/kg}$; 95% CI: 0.5, 1.0) or organic (GM = 0.6 $\mu\text{g/kg}$; 95% CI: 0.5, 0.8) samples. Roxarsone was detected in 20 of 40 conventional samples, 1 of 13 antibiotic-free samples, and none of the 25 organic samples. iAs concentrations in roxarsone-positive samples (GM = 2.3 $\mu\text{g/kg}$; 95% CI: 1.7, 3.1) were significantly higher than those in roxarsone-negative samples (GM = 0.8 $\mu\text{g/kg}$; 95% CI: 0.7, 1.0). Cooking increased iAs and decreased roxarsone concentrations. We estimated that consumers of conventional chicken would ingest an additional 0.11 $\mu\text{g/day}$ iAs (in an 82-g serving) compared with consumers of organic chicken. Assuming lifetime exposure and a proposed cancer slope factor of 25.7 per milligram per kilogram of body weight per day, this increase in arsenic exposure could result in 3.7 additional lifetime bladder and lung cancer cases per 100,000 exposed persons. Conclusions: Conventional chicken meat had higher iAs concentrations than did conventional antibiotic-free and organic chicken meat samples. Cessation of arsenical drug use could reduce exposure and the burden of arsenic-related disease in chicken consumers.